RoleWalk: inductive structural roles for directed graphs from the characteristic function of random walks

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Draft of 01 November 2026 — results and proofs; prose to come

Abstract. Structural roles group nodes by their position in a graph rather than by proximity. RoleWalk describes each node by the characteristic function of its lazy random-walk transition probabilities, sampled at a few frequencies, together with the distribution of the walk's mass over distance layers and, optionally, low-order moments of a probabilistic curvature. The embedding is local, deterministic, reads directions and weights, and needs no training. We prove that it is invariant and local (hence transferable across graphs), that its distances are maximum mean discrepancies, that it is Lipschitz in edge insertions, that it counts triangles, which places it beyond 1-WL, and that distance-regular graphs with equal intersection arrays are its exact limit. Empirically it matches or beats struc2vec, GraphWave, Role2Vec, RESD and ReFeX on role classification at 10–1,000× lower cost, is best on directed roles and on transfer of centralities across graphs, and improves GPS on ZINC when combined with RWSE.

1 Introduction

Claims, each backed by a result below:

1. All of: inductive, directed, weighted, deterministic, online, fast (Table 1).
2. Accuracy at or above the state of the art on role classification at a fraction of the cost (Table 2, Figure 1, Table 11).
3. Only method that uses directions; best on synthetic and real directed roles (Table 4, Table 5).
4. Transfer: a model fitted on one graph applies to another (Table 8, Table 9; Proposition 1).
5. Expressiveness beyond 1-WL with 3 walk steps, on par with 16-step RWSE; exact characterisation of a failure class (Section 4, Table 6).
6. Useful as a structural encoding for graph transformers (Table 10).

2 Related work

Feature-based roles: ReFeX [1], RolX [2]. Walks and word2vec: struc2vec [3], Role2Vec [4]; directed and weighted extensions of struc2vec [5] [TODO: compare if code is available]. Diffusion: GraphWave [6], closest to RoleWalk (Proposition 10). Deep role embeddings: RESD and the RONE benchmark [7] [TODO: exact references]. Structural encodings for GNNs: RWSE [8], used by GPS [9]. Expressiveness: 1-WL [10], [11], [12], CSL [13], BREC [14]. Curvature: Ollivier-Ricci [15], [16]. Kernel mean embeddings and MMD: [17].

3 Method

Walk. $G = (V, E)$ with non-negative weights A (directed or not), $\tilde{D} = \mathrm{diag}((I + A)\mathbf{1})$ and the lazy walk $H = \tilde{D}^{-1}(I + A)$. For a node u and a step w, $p_u^w = e_u^\top H^w$ is the distribution of the walk after w steps. Directed graphs use the out-walk (A) and the in-walk ($A^\top$) and concatenate both embeddings.

Characteristic function. With $f_\theta(x) = e^{i\theta x} - 1$ and a grid $\Theta = \{\theta_1, ..., \theta_m\}$ (log-spaced in $[1, 100]$, $m = 10$),

$$\varphi_u^w(\theta) = \sum_{j \in V} f_\theta\left(p_{uj}^w\right), \quad \theta \in \Theta. \tag{1}$$

Since $f_\theta(0) = 0$, the sum runs over the support of p_u^w only.

Onion layers. With $d(u, j)$ the length of the shortest walk from u to j, $s_u^w(k) = \sum_{j:d(u,j)=k} p_{uj}^w$ for $k = 0, ..., w$: the mass the walk keeps at each distance, which the multiset of probabilities ignores. Supports of earlier powers give the layers, so no extra walk is computed.

Curvature (option). For an edge xy of the symmetrised unweighted graph, $\kappa(x,y) = \sum_z \min\big(m_x(z), m_y(z)\big)$ with m_x the one-step lazy walk from x; κ_x is its mean over the edges at x. Each step adds the mean $\mu_u^w = \sum_j p_{uj}^w \kappa_j$ and the standard deviation σ_u^w of κ under p_u^w. Since a unit of mass that does not overlap moves at least one step, $W_1\big(m_x, m_y\big) \geq 1 - \kappa(x,y)$: κ is an upper bound of Ollivier-Ricci curvature $1 - W_1$, made of probabilities.

One measure, one kernel. Every node j reached at step w is an atom marked by its step and its layer, which gives the *footprint* of the walks from u:

$$\mu_u = \sum_{w=1}^{L} \sum_{j: p_{uj}^w > 0} \delta_{\big(w, p_{uj}^w, d(u,j)\big)}. \tag{2}$$

The embedding is the mean of a single feature map over the footprint, $X_u = \int \Psi \, d\mu_u$, with

$$\Psi(w, a, \ell) = e_w \otimes (\psi(a) \oplus \|\Theta\| a e_\ell), \quad \psi(a) = \big(f_\theta(a)\big)_{\theta \in \Theta}, \tag{3}$$

which gives, for each step $w = 1, ..., L$ ($L = 3$), $\operatorname{Re}\varphi_u^w$, $\operatorname{Im}\varphi_u^w$ and $\|\Theta\| s_u^w$. With curvature the mark is $\big(d(u,j), \kappa_j\big)$ and Ψ adds $\|\Theta\| a(\kappa, \kappa^2)$; we report $\|\Theta\|(\mu_u^w, \sigma_u^w)$, a fixed reparametrisation of these two moments. In all, $2mL + L(L+3)/2(+2L) = 69$ (75) features. The probability block ignores the layer; the layer block is the first-order part of one characteristic function per layer, which is what fixes the factor $\|\Theta\| = \big(\sum_\theta \theta^2\big)^{1/2}$ (Proposition 3). Without it the layers carry 0.02% of the variance and do not change the roles found by k-means (Section 6.3).

Pruning. After each product, entries below $\mathrm{tol} = 10^{-4}$ are dropped; their mass is added back to $\operatorname{Im}\varphi$ to first order ($\theta \cdot \mathrm{mass}$), the same tangent approximation as the layer block. Results in Section 4 are stated for $\mathrm{tol} = 0$.

Roles. Ward clustering with silhouette for the number of roles, or k-means with a given number; fitted centroids label nodes of any graph.

4 Properties

Throughout, L is the number of steps, $B_r(u)$ the subgraph induced by nodes within distance r of u (out- and in-distance for directed graphs), and statements hold for $\mathrm{tol} = 0$.

Proposition 1 (locality and invariance)**.** *X_u is determined by the isomorphism class of the rooted graph $(B_L(u), u)$, and of $(B_{L+2}(u), u)$ with curvature. In particular it does not depend on $|V|$, nodes related by an automorphism get equal embeddings, and a node can be embedded from its neighbourhood alone.*

Proof. $p_{uj}^w = \sum_{u=x_0,\dots,x_w=j} \prod_t H_{x_t x_{t+1}}$ runs over walks of length $w \leq L$ from u. Their nodes lie within distance w, and $H_{xy} = (I + A)_{xy}/\tilde{D}_{xx}$ needs the edges at x, with x within distance $w - 1$: all are in $B_L(u)$. An isomorphism of rooted balls maps these walks bijectively and preserves their weights, so it maps p_u^w to $p_{u'}'^w$ entry by entry; Equation 1 is a sum over j, hence equal. Distances up to L are read inside $B_L(u)$. With curvature, κ_j for j within distance L needs the closed neighbourhoods of j and of its neighbours, inside $B_{L+2}(u)$. □

Coordinates therefore mean the same thing in every graph, which is what transfer (Table 8, Table 9) requires.

Proposition 2 (injectivity)**.** *Let P, Q be finite multisets of reals in $(0, 1]$. If $\sum_{p \in P} f_\theta(p) = \sum_{q \in Q} f_\theta(q)$ for all θ in a set with an accumulation point (e.g. $[1, 100]$), then $P = Q$. For a finite grid, given any countable family of pairs $P \neq Q$, Lebesgue-almost every $\Theta \in \mathbb{R}^m$ separates all of them.*

Proof. $F(\theta) = \sum_P e^{i\theta p} - \sum_Q e^{i\theta q} - (|P| - |Q|)$ is entire and vanishes on a set with an accumulation point, hence everywhere. Exponentials $e^{i\theta\lambda}$ with distinct λ are linearly independent, so each $\lambda > 0$ has the same multiplicity in P and Q (and $|P| = |Q|$, coefficient of $\lambda = 0$). For the second claim, the real zeros of each nonzero entire function F are countable; a countable union of countable sets is null, so almost every θ_1 already separates every pair. □

So with Θ an interval, φ_u^w is exactly the multiset of nonzero transition probabilities of u after w steps. The second claim is qualitative: the separation margin, not its existence, motivates $m = 10$ frequencies across two decades.

Proposition 3 (one kernel on the footprint)**.** $\|X_u - X_v\|^2 = \mathrm{MMD}_K^2(\mu_u, \mu_v)$ *for the positive-definite kernel on marked atoms*

$$K((w,a,\ell),(w',b,\ell')) = \delta_{ww'}\,[k_\Theta(a,b) + \|\Theta\|^2 ab\,\delta_{\ell\ell'}], \tag{4}$$

$$k_\Theta(a,b) = \sum_{\theta\in\Theta}[\cos\theta(a-b) - \cos\theta a - \cos\theta b + 1]. \tag{5}$$

The first term compares probabilities whatever their layer, the second compares masses in the same layer. Moreover, with $t = \max(\Theta)\max(a,b)$*,*

$$\left|k_\Theta(a,b) - \|\Theta\|^2 ab\right| \le \|\Theta\|^2 ab\,t^2(7/12 + t^2/36), \tag{6}$$

so the layer term is the first-order part of $k_\Theta(a,b)\,\delta_{\ell\ell'}$*, the kernel of one characteristic function per layer, and* $\|\Theta\|$ *is the only factor for which the two agree to first order.*

Proof. $X_u = \sum \Psi$ over the atoms of μ_u, so X_u is the kernel mean embedding of μ_u and $\|X_u - X_v\|^2$ is the MMD [17] for $K = \mathrm{Re}\langle\Psi,\Psi'\rangle$, positive definite as an inner product. The blocks e_w and e_ℓ are orthonormal, which gives $\delta_{ww'}$ and $\delta_{\ell\ell'}$; expanding $\mathrm{Re}\big(e^{i\theta a}-1\big)\overline{(e^{i\theta b}-1)}$ gives k_Θ. Unreached nodes have $\Psi = 0$ and do not count. For the bound, fix θ and write the real part as $(\cos\theta a - 1)(\cos\theta b - 1) + \sin\theta a \sin\theta b$. Since $|\cos x - 1| \le x^2/2$, the first product is at most $\theta^4 a^2 b^2/4 \le \theta^2 ab\,t^2/4$. With $\sin\theta a = \theta a + r_a$ and $|r_a| \le \theta^3 a^3/6$, $\left|\sin\theta a \sin\theta b - \theta^2 ab\right| \le \theta^4 ab(a^2+b^2)/6 + \theta^6 a^3 b^3/36 \le \theta^2 ab\,t^2(1/3 + t^2/36)$. Summing over θ gives the bound, and $\|\Theta\|^2 = \sum_\theta \theta^2$ is the coefficient of ab. □

The concatenation is therefore a sum of kernels on one object: a mark-blind term on probabilities and a mark-aware term on masses. The first-order regime $t \ll 1$ holds for hubs, whose walks spread thin, but not for low-degree nodes ($\max(\Theta) = 100$): the linear layer kernel is a choice, $L(L+3)/2$ coordinates instead of $2m(L+1)$ per step, and the full per-layer kernel is not better (Table 13). Further marks enter the same way, as factors of the second term: curvature, or labels on the steps of each path.

Proposition 4 (stability)**.** *Let* G' *be* G *plus an unweighted edge* ab*, with* $d = \min(d_a, d_b)$ *the smaller endpoint degree in* G*. For every node* u *and step* w*,*

$$\left\|\varphi_u^w(G') - \varphi_u^w(G)\right\|_2 \le w\|\Theta\|_2 \cdot \frac{2}{d+2}, \tag{7}$$

and the left side is 0 if u *is at distance* $\ge w$ *from both* a *and* b*.*

Proof. $|e^{i\theta x} - e^{i\theta y}| \le \theta\,|x-y|$, so $\left\|\varphi_u^w(G') - \varphi_u^w(G)\right\|_2 \le \|\Theta\|_2 \|p'^w_u - p^w_u\|_1$. Telescoping, $H'^w - H^w = \sum_{t=0}^{w-1} H'^t(H'-H)H^{w-1-t}$; stochastic matrices have $\|\cdot\|_\infty = 1$, so each row of the difference has ℓ_1 norm at most $w\|H'-H\|_\infty$. Only rows a and b of H change; row a goes from d_a+1 entries $1/(d_a+1)$ to d_a+2 entries $1/(d_a+2)$, an ℓ_1 change of $2/(d_a+2)$. Row u of $H'^t(H'-H)$ is nonzero only if a walk of $t \le w-1$ steps from u reaches a or b. □

Insertions at high-degree nodes barely move embeddings, and the effect is confined to the $(L-1)$-ball. On 200 random insertions (powerlaw-cluster graphs, 60 nodes) the largest observed move is 0.52 of the bound (`theory_checks.py`).

Proposition 5 (degree and triangles)**.** $s_u^1(0) = 1/(d_u+1)$ *in an unweighted graph. If* G *is* d*-regular,* $s_u^3(0) = (H^3)_{uu} = (1+3d+2t_u)/(d+1)^3$*, with* t_u *the number of triangles at* u*. Hence RoleWalk with* $L \ge 3$ *separates 1-WL-equivalent graphs, e.g. two triangles and a hexagon, or any two* d*-regular graphs with different triangle counts.*

Proof. $(I+A)^3 = I + 3A + 3A^2 + A^3$ with $(A)_{uu} = 0$, $(A^2)_{uu} = d$, $(A^3)_{uu} = 2t_u$, and $H = (I+A)/(d+1)$. 1-WL gives every node of two d-regular graphs of the same size the same colour. □

Proposition 6 (relation to RWSE)**.** *The distance-0 layer* $s_u^w(0) = (H^w)_{uu}$ *is the return probability of the lazy walk, i.e. RWSE [8] for* H*, up to* L *steps. On a* d*-regular graph,* $\left\{(H^w)_{uu}\right\}_{w\le L}$ *and the simple-walk RWSE* $\left\{(P^k)_{uu}\right\}_{k\le L}$*,* $P = A/d$*, determine each other.*

Proof. $H = (I+dP)/(d+1)$, so $(H^w)_{uu} = (d+1)^{-w}\sum_{k=0}^{w}\binom{w}{k} d^k (P^k)_{uu}$: a triangular system with nonzero diagonal $d^w/(d+1)^w$. □

On irregular graphs the two walks differ and neither encoding contains the other; RoleWalk adds the other layers and the full distribution of transition probabilities.

Proposition 7 (distance-regular graphs are the limit)**.** *Let G, G' be distance-regular graphs with the same intersection array. Then all nodes of G and G' get the same embedding, for every L, with or without curvature. This covers strongly regular graphs with equal parameters (diameter 2).*

Proof. Let A_i be the distance-i matrices. Distance-regularity gives $AA_i = b_{i-1}A_{i-1} + a_iA_i + c_{i+1}A_{i+1}$ with coefficients from the intersection array [18], so by induction $(I + A)^w = \sum_i \alpha_{w,i}A_i$ with $\alpha_{w,i}$ depending only on the array. As $H = (I + A)/(k + 1)$, $p^w_{uj} = \alpha_{w,d(u,j)}/(k + 1)^w$: each row has value $\alpha_{w,i}/(k + 1)^w$ with multiplicity k_i, the number of nodes at distance i, fixed by the array. So φ^w_u and $s^w_u(i) = k_i\alpha_{w,i}/(k + 1)^w$ are the same for all nodes of both graphs. For an edge xy, $\kappa(x, y) = |N[x] \cap N[y]| /(k + 1) = (a_1 + 2)/(k + 1)$, a constant, so μ^w_u is constant and $\sigma^w_u = 0$. □

Checked on Shrikhande / 4×4 rook, Chang / $T(8)$ and the diameter-3 pair Doob $D(1, 1)$ / Hamming $H(3, 4)$: largest difference between any two nodes $\leq 3 \cdot 10^{-7}$ (float32). This explains the 0/20 distance-regular pairs of BREC (Table 7); RWSE, which is constant on these graphs for the same reason, fails them too.

Proposition 8 (locality limits graph-level separation)**.** *Let A and B be a 40-cycle with pendant nodes attached at cycle positions $\{0, 15\}$ and $\{0, 20\}$. They are not isomorphic and 1-WL separates them, but RoleWalk with $L = 3$ (with curvature) does not.*

Proof. The pendants are more than $2(L + 2)$ apart in both graphs, so every rooted ball $B_{L+2}(u)$ contains at most one pendant and is fixed by u's distance to it; both graphs have the same multiset of balls. Conclude by Proposition 1. Colour refinement propagates along the cycle and eventually sees the distance between pendants (checked with 40 rounds). □

Empirical comparison with 1-WL at equal depth. A walk of L steps sees degrees up to distance $L - 1$, as does 1-WL after L rounds from uniform colours. Over all 6,781 nodes of the 996 connected graphs with at most 7 nodes, the partition induced by RoleWalk ($L = 3$) refines that of 3-round 1-WL: no two nodes that 1-WL separates share an embedding, and RoleWalk splits 571 of the 1-WL classes (embeddings grouped at a sup-distance below 10^{-4} of the largest coordinate; `theory_checks.py`).

Conjecture 9. *For every L, the partition of nodes (across all graphs) induced by RoleWalk with L steps and Θ an interval refines the partition of L-round 1-WL.*

Together with Proposition 5 and the propositions above: at equal depth RoleWalk is strictly finer than 1-WL on all graphs up to 7 nodes, while being local, which 1-WL with unbounded rounds is not.

Proposition 10 (relation to GraphWave)**.** *GraphWave [6] embeds u by $\chi_u(t) = \frac{1}{n}\sum_j e^{it\Psi_{ju}}$ with $\Psi = Ug_s(\Lambda)U^\top$ the heat kernel at a scale s chosen from the spectrum of each graph. Then $n(\chi_u(t) - 1) = \sum_j f_t(\Psi_{ju})$: Equation 1 with the heat kernel in place of H^w. χ_u depends on n and on s, so the same rooted neighbourhood gets different coordinates in graphs of different sizes or spectra.*

Proof. $\sum_j\big(e^{it\Psi_{ju}} - 1\big) = n\chi_u(t) - n$. Ψ has full support on a connected graph, so Proposition 1 fails; $1/n$ and s are graph-level. □

This is why GraphWave transfers at chance level (Table 9), while RoleWalk keeps its accuracy.

Proposition 11 (cost)**.** *With pruning, each row of each power keeps at most $1/\operatorname{tol}$ entries, so step w costs $O\Big(\sum_u \sum_{j\in \operatorname{supp} p_u^{w-1}}\big(d_j + 1\big)\Big)$ with $|\operatorname{supp} p_u^{w-1}| \leq \min(|B_{w-1}(u)|, 1/\operatorname{tol})$; the characteristic function adds $O(m \cdot \operatorname{nnz})$ and the layers L element-wise products. Curvature costs $O\big(\sum_u d_u^2\big)$. Embedding a set S touches only $B_L(S)$ ($B_{L+2}(S)$ with curvature).*

Proof. Entries are non-negative and each row sums to at most 1, so at most $1/\operatorname{tol}$ of them are $\geq \operatorname{tol}$. The rest follows from sparse products and Proposition 1. □

5 Experiments

All CPU timings on one core. Classification: logistic regression on standardised embeddings, 50 stratified 80/20 splits. Best in bold, second underlined. Code and CSV files in `benchmarks/`.

Table 1: Properties. Checked against each paper; "~" partial.

	Transfer	Directed	Weighted	Time (USA)	Online	Deterministic
RoleWalk	yes	yes	yes	0.2 s	yes	yes
struc2vec	no	no[a]	no[a]	564 s	no	no
GraphWave	no (P. 10)	~	yes	18 s	no	yes
Role2Vec	~	yes[b]	no	10 s	~	no
RESD	no	no	no	8 s	no	no
ReFeX	~	~	~	0.1 s	~	yes
RWSE	yes	~	yes	1.7 s	yes	yes

[a] struc2vec++ [5] extends it. [b] In the paper; the reference implementation is undirected. [TODO: verify RESD, ReFeX, RWSE rows]

Table 2: Role classification accuracy. struc2vec is unseeded: mean ± std over 5 runs. `roles.csv`, `struc2vec_runs.csv`.

	Brazil	Europe	USA	Reality-call	Actor	Film*
RoleWalk	0.743	0.583	0.657	**0.860**	**0.487**	0.569
RoleWalk + curv.	0.741	**0.596**	**0.666**	**0.860**	0.485	0.581
struc2vec	0.705 ± .022	0.561 ± .020	0.596 ± .004	–	–	–
RESD	0.728	0.544	0.653	0.857	0.476	0.563
GraphWave	**0.769**	0.561	0.587	**0.860**	0.481	–
ReFeX	0.691	0.576	0.622	0.851	0.478	0.579
Role2Vec	0.476	0.366	0.562	0.844	0.399	**0.743**
RWSE	0.711	0.572	0.517	0.855	0.363	0.450
Degree	0.756	0.554	0.567	0.858	0.476	0.478

* A node-type task (Section 6.4). "–": too slow or out of memory.

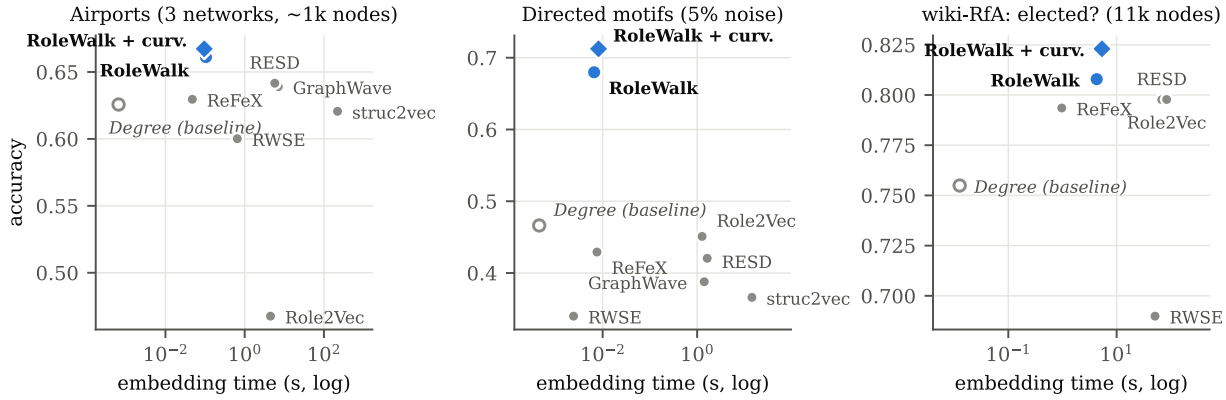


Figure 1: Accuracy against embedding time (one core, log scale), every method that runs on each benchmark. Airports: mean over Brazil, Europe and USA (struc2vec: mean of 5 runs). Directed motifs: 5% rewired edges, mean of 3 seeds. wiki-RfA: task A of Table 5. Hollow marker: degree, a trivial baseline rather than an embedding. RoleWalk is the only method in the top-left corner of every panel. `speed_figure.py`.

Table 3: Wikipedia talk networks (RONE): precision@10 for bots and admins, mean over networks. `roles.csv`.

	Bots	Admins
RoleWalk	0.129	0.324
struc2vec (3 smallest networks)	**0.146**	**0.391**
GraphWave	0.115	0.298
RESD	0.114	0.254
ReFeX	0.101	0.299
RWSE	0.118	0.297
Degree	0.075	0.296

Mixed. Networks are distributed undirected. [TODO: directed versions from SNAP; per-network table]

Table 4: Directed roles on synthetic graphs (directed motifs hung from a directed cycle; several roles coincide once directions are dropped). 3 seeds; noise = share of edges rewired. `directed.csv`.

	Acc. 0%	Acc. 5%	Acc. 10%	ARI 5%	mAP 5%
RoleWalk + curv.	**0.989**	**0.712**	**0.547**	**0.381**	**0.625**
RoleWalk	0.908	0.680	0.510	0.366	0.574
RoleWalk, undirected	0.767	0.560	0.420	0.351	0.479
Role2Vec	0.730	0.451	0.368	0.148	0.295
ReFeX	0.532	0.429	0.431	0.342	0.356
GraphWave	0.586	0.388	0.383	0.317	0.416
RESD	0.547	0.420	0.423	0.253	0.348
struc2vec	0.510	0.366	0.277	0.141	0.260
In/out degree	0.516	0.466	0.405	0.310	0.349

Table 5: Real directed roles: Wikipedia requests for adminship [19] (11,377 users, 188k votes voter → candidate). A: elected among the 3,494 candidates. B: voter / failed candidate / admin. `roles_rfa.csv`.

	A: elected	B: 3 classes	Time
RoleWalk + curv.	**0.823**	**0.943**	5 s
RoleWalk	0.808	0.938	4 s
RoleWalk, undirected	0.791	0.863	
RoleWalk + curv., undirected	0.814	0.900	
RESD	0.798	0.843	68 s
Role2Vec	0.798	0.857	82 s
ReFeX	0.794	0.847	1 s
In/out degree	0.755	0.922	0 s
RWSE (out-walk)	0.690	0.826	51 s

struc2vec and GraphWave do not run at this size.

Table 6: Expressiveness: pairs told apart. Every set is 1-WL-indistinguishable. No false difference on random relabellings. `expressiveness.csv`.

	Atlas ≤ 7 (20)	Regular (3,181)	CSL (10 cl.)	Decalin / bicyc.	$2C_3$ / C_6	SRG pair*
RoleWalk (3 steps)	**20**	**3,181**	6	yes	yes	no
RoleWalk (5 steps)	**20**	**3,181**	**10**	yes	yes	no
RoleWalk, no layers	16	3,180	6	yes	yes	no
RWSE (16 steps)	**20**	3,180	**10**	yes	yes	no
Degree (= 1-WL)	0	0	1	no	no	no

* Shrikhande / 4×4 rook, beyond 3-WL; not separable by Proposition 7.

Table 7: BREC [14] (400 pairs).

	Basic (60)	Regular (100)	Extension (100)	CFI (100)	4-vertex (20)	Dist.-reg. (20)	Total
RoleWalk (3 steps)	60	50	97	0	0	0	207
RoleWalk (5 steps)	60	50	100	3	0	0	213
RoleWalk, no layers	58	48	97	0	0	0	203
RWSE (16 steps)	60	50	100	4	0	0	**214**

Distance-regular: 0 by Proposition 7. The 50 regular pairs missed are exactly the 50 strongly regular pairs, all with equal parameters (checked).

Table 8: Centrality transfer: a gradient-boosting regressor from embedding to centrality is fitted on one graph and evaluated (Spearman) on another graph of the same family and size (other seed) (Barabási-Albert, powerlaw-cluster, Watts-Strogatz, SBM, random geometric, motifs). `centrality_transfer.csv`.

	RoleWalk	+ curv.	ReFeX†	RWSE	GraphWave	Degree
Betweenness	0.901	**0.905**	0.844	0.708	0.723	0.634
Closeness	**0.866**	0.864	0.684	0.732	0.597	0.428
Clustering	**0.873**	0.870	0.852	0.803	0.392	0.212
k-core	0.747	0.748	**0.935**	0.660	0.566	0.692
Eigenvector	**0.656**	0.649	0.492	0.531	0.462	0.428
Harmonic	**0.918**	0.916	0.782	0.784	0.699	0.500
PageRank	**0.990**	**0.990**	0.968	0.864	0.857	0.927
Mean	**0.850**	0.849	0.794	0.726	0.614	0.546

† Only where its number of features matches across graphs.

Table 9: Role transfer between airport networks: classifier fitted on one network, tested on another (6 ordered pairs, accuracy). Labels are activity quartiles, close to degree. `role_transfer.csv`.

Degree	RoleWalk	+ curv.	RoleWalk, no layers	RWSE	GraphWave
0.542	**0.488**	0.470	0.445	0.375	0.249 (chance)

Table 10: Structural encodings for GNNs on ZINC (test MAE at the best validation epoch). Top: GINE, 4 layers, ≈230k parameters, 300 epochs, 3 seeds. Bottom: GPS [9] protocol, 10 layers, ≈470k parameters, 500 epochs, 2 seeds. `pe_zinc_v13.csv`, `pe_zinc_gps.csv`.

GINE	Test MAE	
none	0.202 ± 0.010	
RWSE	0.107 ± 0.004	
RoleWalk	0.108 ± 0.008	
RoleWalk + RWSE	0.104 ± 0.006	
RoleWalk + curv. + RWSE	**0.095 ± 0.002**	
GPS	**Test MAE**	**Val. MAE**
RWSE (published: 0.070 ± 0.004)	0.077 ± 0.006	0.103 ± 0.000
RoleWalk + curv.	0.076 ± 0.001	0.092 ± 0.003
RoleWalk + curv. + RWSE	**0.073 ± 0.001**	**0.089 ± 0.005**

[TODO: GPS seeds 2-3 running; Peptides (LRGB)]

Table 11: Embedding time against size, one thread, defaults (tol 10^{-4}, 3 steps, layers). `scalability.csv`.

Nodes	10 k	30 k	100 k	300 k	1000 k
Barabási-Albert ($m = 3$)	1.11 s	3.72 s	15.16 s	63.8 s	254.56 s
+ curvature	1.19 s	3.87 s	16.61 s	55.29 s	239.92 s
Watts-Strogatz ($k = 6$)	0.12 s	0.43 s	1.46 s	5.4 s	19.93 s
+ curvature	0.13 s	0.47 s	3.43 s	5.83 s	28.17 s

Peak memory 6.6 GB at a million nodes (blocks of 50k rows). For comparison, at 1,190 nodes (USA airports): struc2vec 564 s, GraphWave 18 s, Role2Vec 10 s, RESD 8 s; struc2vec and GraphWave do not run on wiki-RfA (11k nodes).

6 Analysis

6.1 Summaries of the walk distribution

Table 12: Same budget of features, replacing the characteristic function. Airports: mean accuracy; Transfer: airport transfer; Motifs: synthetic accuracy.

	Airports	Transfer	Motifs (varied / houses)	Time 20k
Characteristic function	0.656	0.445	**0.744** / 0.733	6.5 s
Laplace transform	**0.657**	0.345	0.614 / 0.717	18 s
Dyadic histogram (Haar)	0.631	0.353	0.614 / 0.725	15 s
Power sums (Rényi)	0.653	0.420	0.647 / 0.692	9.7 s
Top-10 probabilities	0.652	**0.512**	0.679 / **0.758**	3.5 s

6.2 Geometry: moments rather than integrated characteristic functions

Table 13: Ways to add local geometry. Anything that depends on graph size (degree marks, counting bounds) hurts transfer.

	Airports	Transfer	Motifs (varied / houses)	Directed acc.
No geometry	0.656	0.445	0.744 / 0.733	0.678
+ onion layers (moments)	0.661	0.488	0.744 / 0.758	0.682
+ layers, one CF per layer	0.654	0.424	0.751 / 0.742	0.695
+ degree marks (joint CF)	0.65–0.67	0.34–0.36	–	–
+ curvature, counting bound	0.667	0.388	**0.754** / **0.767**	0.721
+ curvature, probabilistic (moments)	**0.669**	0.470	0.725 / 0.742	0.712
+ curvature, exact Ollivier-Ricci	0.662	**0.494**	0.746 / 0.750	**0.743**
+ curvature, integrated CF	0.663	0.400	0.751 / 0.750	0.697

The probabilistic bound is kept over exact Ollivier-Ricci for cost (no optimal transport) at a small loss.

Marks on the steps of a path rather than on its end were also tried: the parity of steps inside a layer (closing odd cycles), obtained for free from the signed walk $\tilde{D}^{-1}(A - I)$ since $\#\text{intra} \equiv \#\text{moves} - d(u, j) \pmod 2$, its exact count, and the order of moves out and in, as products in the quaternion group Q_8. On the 6,781 nodes of connected graphs with at most 7 nodes, none separates a pair of nodes that X does not, and none improves airport accuracy (`intra_layer_study.py`, `quaternion_study.py`): with commuting labels the product along a path only depends on its end layer, and short cycles already change the multiset of probabilities.

6.3 Kernel scaling

In raw embeddings $\theta = 100$ carries 53% of the variance; unscaled layers carry 0.02% and leave k-means roles unchanged. A grid of 48 configurations (frequency weighting, range, steps, layers), scored on synthetic and real networks with equal weight (`kernel_grid.csv`): scaling the layers by $\|\Theta\|$ is what matters (score 0.274 $\rightarrow$ 0.323), the frequency weighting barely does, and the defaults win whichever dataset is left out. Spacing matters less than range (`theta_scheme.csv`): on $[1, 100]$, log-spaced frequencies score 0.323 against 0.316 for uniform ones, with the same airport accuracy (0.661 / 0.662) and transfer (0.488 / 0.485). A narrower range ($[1, 30]$) helps real networks and transfer (0.515) but loses on synthetic motifs (score 0.315). Log spacing is kept because probabilities scale multiplicatively ($p \approx 1/d^w$): a change of degree rescales p, which shifts a log grid along itself, while a uniform grid puts 9 of its 10 frequencies above 10.

6.4 Film is a node-type task

Table 14: Film (RONE). Class 3 is an independent set in a separate block of node ids.

	Class 3 vs rest	Within classes 0–2
Role2Vec	**0.946**	0.637
RoleWalk + curv.	0.740	<u>0.634</u>
ReFeX	<u>0.745</u>	0.627
Degree	0.536	0.623
Majority class	0.593	0.623

Classes 0–2 are not predictable by any method; edge homophily is 0.28, below the 0.33 of random labels. Role2Vec's lead comes from walk co-occurrence, i.e. position, which an automorphism-invariant embedding cannot capture by Proposition 1. Film should not rank role methods.

7 Limitations

- Walk statistics cannot separate distance-regular graphs with equal arrays (Proposition 7), CFI pairs, or graphs whose balls coincide (locality).
- Middle-tier roles on real networks often form a continuum; per-node stability should be reported next to roles.
- Degree-like labels (airports) favour trivial baselines.
- One real labelled directed benchmark.
- GPS on ZINC: 2 seeds; the test gain is within published seed noise, clearer on validation.

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